Aerospace Mechanics and Controls

Planes, Rockets, Quadcopters, Satellites and everything in between

Carlos Montalvo
University of South Alabama Mobile, AL[1]

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[1]William B. Burnsed Jr. Department of Mechanical, Aerospace and Biomedical Engineering, Associate Professor

Website: Facility for Aerospace Systems and Technologies (FAST) Lab

0.1 Current Edition

This textbook was last updated on August 18, 2026. The latest pdf can be found on Github while the most up to date web based version is on Github Pages[14].

A note on AI: Note that many sections have been written with the help of Github's Copilot [1] and even Google's Gemini [2]. Originially their were plans to properly cite each section that was written with the help of AI but the authors lost track given it's seamless integration with VScode and Android smart phones as well as any browser at this point. As such, it is important to note that after September of 2025 it is safe to assume that most if not all sections were written with the help of AI. The authors are aware of the ethical implications of this issue however are rigorously check all sections and text to ensure that the content is correct and free of errors.

0.2 Manuscript Changes

1. June 10th, 2020 - Magnetic Field Model section updated to reflect the difference between the East North Vertical and the North East Down reference frames. The Figure showing the magnetic field for an example Low Earth Orbit has also been update. Also added this page for manuscript changes and the following pages that list where this file can be found.
2. December 10th, 2020 - Moved to public Github Repo separate from MATLAB
3. December 30th, 2020 - Moved papers.bib to parent directory Added datetime to title page. Added section headers to RC aircraft design. Wrote text all the way through the airfoil selection. Added a references section.
4. December 31st, 2020 - Finished RC aircraft section.
5. September 30th, 2021 - Renamed report to Aerospace Mechanics and Controls
6. December 21st, 2021 - Moved CubeSAT abstract to introduction on CubeSATs. Imported entire Aircraft Mechanics textbook into here
7. December 22nd, 2021 - Added a section on GNC design for CubeSats. Added acknowledgements section.
8. June 2nd, 2022 - Added some items to changes needed and fixed two references
9. July 30th, 2022 - Included a derivation of direct measurement of Euler Angles using an IMU.
10. July 31st, 2022 - Added GPS coordinate conversion to cartesian coodinates as well as heading angle and speed estimation from GPS.
11. October 27th, 2022 - Added computation of lat, lon, alttiude to ECI frame.
12. March 20th, 2024 - Added a Current Edition section above Manuscript Changes. Also added color to hyperlinks
13. May 7th, 2024 - Added a battery sizing section to the aircraft design chapter
14. July 20th, 2024 - separated sections into separate files...Also started adding the quacopter aerodynamic model
15. July 22nd, 2024 - Finished the quadcopter aerodynamics section and made a quick edit to the GNC aircraft PID control scheme
16. December 23rd, 2024 - Added a better description to the gravity model to explain the vector notation of the equations
17. January 6th, 2025 - A title change was performed for the GPS to Cartesian coordinate transform. Also made orbital elements its own section.
18. January 9th, 2025 - Edited the position of different sections so that it's ready for the linear controls section
19. May 13th, 2025 - Added drag equation to aerodynamics

20. September 7-8th, 2025 - Used Github's copilot on the Mobile App to create the first order example of a differential equation. Also used Gemini as well to create the second order example. I also added a citation for Gemini in the papers.bib file. I also edited the foreword to include a note on AI since from here on out I will use it to help write many sections of this manuscript.
21. September 10th, 2025 - Added the equations of motion for a second order mass spring damper system.
22. September 28th, 2025 - Moved the section on numerical integration to be before the LTI section since it makes more sense to have it there. Also added the second order solution showing underdamped, overdamped and critically damped solutions.
23. September 29th, 2025 - Added the time response section for first order systems and finished the derivation of first and second order systems.
24. September 30th, 2025 - added more section headers to the LTI section to make it easier to navigate.
25. October 10th, 2025 - Split the LTI Controls section into its own file and started writing the feedback control section.
26. October 11th, 2025 - Added the attitude dynamics of a satellite/quadcopter section. Also rearranged the sections overall
27. October 12th, 2025 - Finished the rocket dynamics equations and started on the stability section.
28. October 13th, 2025 - Added the pitch dynamics of an aircraft
29. October 14th, 2025 - Finished the dynamics section of LTI systems and am working on the time response section now
30. October 15th, 2025 - Added the week by week schedule for the project based learning section
31. October 18th, 2025 - Added the code for simulating the mass spring damper system
32. October 19th, 2025 - Added stability sections for mass spring damper and the car velocity/position
33. October 20th, 2025 - Added the time response and stability sections for the quadcopter/satellite attitude dynamics
34. October 21st, 2025 - Split the lti section again so that's it's easier to read. Also finished the pitch dynamics of an aircraft.
35. October 22nd, 2025 - Added the rocket response and stability section.
36. October 24th, 2025 - Added the pendulum response and stability section
37. December 5th, 2025 - Fixed the aircraft dynamics equations as there was a sign error. Updated changes needed onto Github.
38. August 18th, 2026 - Started the conversion from LaTeX to PreTeXt.

0.3 Changes Needed and Future Project Ideas

Needed changes and Future Project Ideas are now tracked on Github

Contents

Chapter 1

Conclusions

Turns out I never wrote a conclusion section for this book. I guess I will get back to this later.

Chapter 2

Acknowledgements

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